Exact-Root Certification of Finite-Error Ordering in Quantum Control

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Abstract

Quantum controls that agree at the nominal endpoint and in their complete first-order response to the declared global error coordinates need not have the same finite-error robustness. We study this residual implementation dependence in a serialized two-atom Rydberg model with twenty-four control phases and three global error coordinates.

Our main result is a computer-assisted exact-root finite-error ordering theorem. For a frozen twelve-path cohort, 192-bit Krawczyk inclusions certify locally unique roots at which the projective output state and its complete first-order response exactly match the reference. Outward-rounded Arb propagation of the resulting root boxes through a six-axis finite-error functional produces twelve pairwise disjoint performance intervals. Their order agrees with a ranking frozen before direct propagation, thereby certifying all 66 unordered path comparisons.

A separate mechanism certificate, not used in the direct theorem, partially recovers this ordering without direct finite-error propagation. An order-thirty zero-error Taylor jet, equipped with Cauchy-alias and Dyson-tail enclosures, certifies 52/66 exact-root box comparisons with no certified reversal; its quartic-only boundary resolves 34/66 comparisons at the serialized frozen decimal points. On a separate twenty-path prospective cohort, a frozen quartic response contraction predicts the held-out ranking above the predeclared threshold $\rho_{\text{Spearman}} \geq 0.95$; recorded numerical-architecture samples give $\rho \approx 0.989$–0.994. Its symmetric tensor representation is verified to transform covariantly under nonorthogonal changes of error coordinates.

The prospective predictor is not a premise of the ordering theorem. All conclusions are conditional on the declared finite-dimensional model and error functional and provide no hardware, model-discrepancy, global-fibre, or many-body certification.

1 Introduction

Programmable neutral-atom systems expose a rich pulse-level control space, including Rabi amplitude, phase, detuning, atomic geometry, and Rydberg-mediated interactions [1, 2, 3]. This flexibility permits many control schedules to realize the same nominal task. It also creates an identification problem: nominally equivalent implementations can respond differently when calibration errors are finite.

Endpoint equivalence does not resolve this ambiguity. Even matching the complete first-order output-state response to the relevant error coordinates may leave a family of implementations with different higher-order behaviour. The resulting question is not whether leading errors can be cancelled, but whether the remaining finite-error ordering can be inferred before those finite-error outcomes are evaluated.

Composite-pulse and dynamically-corrected-gate constructions suppress designated low-order error terms [4, 5, 6, 15, 16, 18]; filter-function methods characterize the frequency-domain noise susceptibility of a fixed control [17]; quantum geometric approaches endow state and parameter manifolds with the quantum geometric tensor and its curvature [19]; and control-landscape theory analyses the critical-point structure of control objectives [13, 14]. These methods typically optimize or cancel designated low-order sensitivities of a control. What remains unresolved by these viewpoints is whether controls that are exactly indistinguishable at the endpoint and throughout their complete first-order projective response can nevertheless admit a rigorously certifiable finite-error ordering. Here we condition on that strong equivalence relation—candidate controls must share both the nominal output state and the complete first-order output-state response to global amplitude, detuning, and interaction errors—and study the finite-radius ordering that remains on the implementation fibre, together with its rigorous certification. We then ask:

> *Within the remaining implementation fibre, can a zero-error geometric quantity prospectively predict finite-error robustness, and can that ordering be certified at locally unique projectively matched controls?*

This formulation separates two logically distinct tasks. The first is prediction: a local quantity must be fixed before held-out finite-error evaluation and must rank previously unseen paths. The second is certification: numerical matching must be replaced by exact-root existence and uniqueness, and the finite-error ordering must be enclosed by outward-rounded interval arithmetic, in the validated-numerics tradition of computer-assisted proof [11, 10, 12]. High correlation alone does not provide the second conclusion.

We study a two-atom Rydberg model with twenty-four piecewise-constant phase variables. The numerical matching Jacobian has rank sixteen, leaving an eight-dimensional numerical tangent null space that organizes the candidate implementation fibre.

1.1 Main results

Main theorem (informal statement). Consider the frozen twelve-path cohort, the serialized finite-dimensional two-atom Hamiltonian, the declared transverse phase boxes, and the six-axis mean finite-error functional. For each declared control, a strict Krawczyk inclusion certifies a locally unique root exactly matched in projective output state and first projective response. Direct 192-bit outward-rounded Arb propagation over the twelve certified root boxes produces pairwise disjoint performance intervals whose orientations agree with the pre-outcome frozen ordering for all $\binom{12}{2} = 66$ unordered path pairs. The formal statements and computer-assisted proofs are given in Theorems 1 and 2. This result is conditional on the serialized model and declared error functional; it is not a claim about hardware, model discrepancy, global fibre topology, or many-body scaling.

Mechanism proposition (informal statement). Over the same certified Krawczyk boxes, propagation of the zero-error Taylor jet through order thirty, including formal Cauchy-alias and analytic tail enclosures, certifies 52/66 pairwise comparisons with no reversed certified pair. This order-thirty calculation is a separate, computationally distinct mechanism certificate. It is not a premise or computational step in the direct proof of the main theorem. The formal statement appears in Proposition 2.

Reading guide. Five quantitative results play different logical roles and are not interchangeable:

34/66	quartic-only serialized-control boundary (frozen decimal points)
66/66	frozen-point order-30 numerical diagnostic (point-valued)
52/66	exact-root order-30 mechanism certificate (box-valued)
66/66	direct exact-root finite-error certificate (box-valued, main result)
$\rho_{\text{Spearman}} \geq 0.95$	prospective evidence (separate cohort, not a premise)

The two 66/66 entries are distinct objects: the frozen-point order-30 figure is a *point-valued* numerical diagnostic at the frozen decimal points, whereas the direct exact-root figure is a *box-valued* certificate over the certified root boxes and is the main theorem. The prospective and formal results use disjoint frozen cohorts; the former is not a premise of the latter.

The remainder of the paper follows the logical development from prospective prediction to exact-root certification. Section 2 fixes the model and the benchmark error law; Section 3 defines exact projective response matching and proves a model-independent quadratic-degeneracy lemma; Section 4 introduces the prospective quartic predictor; Section 5 constructs its covariant tensor form; and Section 6 certifies locally unique exact projective matched roots and proves the direct finite-error ordering theorem, together with the separate order-thirty mechanism certificate. The precise model boundary and the next certification theorem are stated in Section 9.

2 Two-atom control model

2.1 Hamiltonian

We use the ordered basis

$$\{|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle\}.$$

Let

$$X_\Sigma = X \otimes I + I \otimes X, \qquad Y_\Sigma = Y \otimes I + I \otimes Y,$$

$$N_\Sigma = n \otimes I + I \otimes n, \qquad N_{rr} = n \otimes n,$$

where $n = |r\rangle\langle r|$. During segment j, the Hamiltonian is

$$H_j(\epsilon) = \frac{\Omega}{2}(1+\epsilon_\Omega)\big(\cos\phi_j\, X_\Sigma + \sin\phi_j\, Y_\Sigma\big) - \Omega\,\epsilon_\Delta\, N_\Sigma + V(1+\epsilon_V)\, N_{rr}. \tag{1}$$

The dimensionless error vector is

$$\epsilon = (\epsilon_\Omega, \epsilon_\Delta, \epsilon_V).$$

The nominal parameters are

$$\Omega = 2\pi \text{ rad}/\mu\text{s}, \qquad V = 4\Omega,$$

with 24 segments of duration

$$\tau = 0.1\ \mu\text{s}, \qquad T = 24\tau = 2.4\ \mu\text{s}.$$

The nominal interaction is represented as $V = C_6/R^6$. Using the frozen Pulser 1.8.0 `DigitalAnalogDevice` value $C_6 = 5420158.53\ \text{rad}\,\mu\text{m}^6/\mu\text{s}$ and the declared $V = 4\Omega = 8\pi\ \text{rad}/\mu\text{s}$ gives

$$R = (C_6/V)^{1/6} = 7.74394075\ldots\ \mu\text{m}.$$

This distance is a serialized model parameter rather than a hardware-calibrated measurement.

The two-atom model is not intended as a many-body approximation. It is a minimal interacting Rydberg setting in which drive-amplitude, detuning, and interaction-calibration errors coexist. A single-atom model has no nontrivial N_{rr} response and therefore cannot represent the three-direction matched response problem studied here.

For a control path

$$\gamma = (\phi_1, \ldots, \phi_{24}) \in \mathbb{T}^{24},$$

the output state is

$$|\psi_\gamma(\epsilon)\rangle = U_{24}(\epsilon)\cdots U_2(\epsilon)U_1(\epsilon)|gg\rangle, \qquad U_j(\epsilon) = \exp[-i\tau H_j(\epsilon)], \tag{2}$$

with time-ordered application from segment 1 to segment 24. All primary calculations use exact matrix exponentials for the piecewise constant, finite-dimensional model. A pulse-level emulator was used in an earlier two-atom cross-check, but the formal certificate in this work is independent of that emulator and does not invoke a cloud backend. Throughout, *formal certificate* denotes an outward-rounded interval certificate computed in Arb/ACB ball arithmetic, not a proof-assistant formalization.

2.2 Loss and held-out error law

Let γ_{ref} denote the reference schedule and define

$$|\psi_{\mathrm{ref}}\rangle = |\psi_{\gamma_{\mathrm{ref}}}(0)\rangle.$$

Two infidelities are used. For the extraction of local response coefficients, the target is the path-local nominal state,

$$\mathcal{I}^{\mathrm{loc}}_\gamma(\epsilon) = 1 - |\langle\psi_\gamma(0)|\psi_\gamma(\epsilon)\rangle|^2. \tag{3}$$

For held-out and direct finite-error comparisons, the target is the common reference state,

$$\mathcal{I}^{\mathrm{ref}}_\gamma(\epsilon) = 1 - |\langle\psi_{\mathrm{ref}}|\psi_\gamma(\epsilon)\rangle|^2. \tag{4}$$

At an exact response-matched root, $[\psi_\gamma(0)] = [\psi_{\mathrm{ref}}]$ projectively, so the two definitions coincide; for numerically matched prospective cohorts, they differ only within the declared matching tolerance. Unless stated otherwise, $\mathcal{I}^{\mathrm{ref}}_\gamma$ below denotes (4).

The held-out error design contains the six signed axial errors

$$E = \{\pm 0.06\, e_\Omega,\ \pm 0.04\, e_\Delta,\ \pm 0.05\, e_V\}. \tag{5}$$

These axis amplitudes define a dimensionless benchmark error law for certificate comparison; they are not inferred from hardware calibration data. The primary performance functional is the mean

$$\bar{\mathcal{I}}_\gamma = \frac{1}{6}\sum_{\epsilon\in E}\mathcal{I}^{\mathrm{ref}}_\gamma(\epsilon). \tag{6}$$

Worst-case infidelity is retained as a secondary diagnostic. The present predictors do not reliably rank the worst case, and no worst-case theorem is claimed.

3 Matched implementation fibre

3.1 Endpoint and first-order constraints

At zero error, remove the unobservable global phase by the path-local horizontal projector

$$P_{\gamma,\perp} = I - |\psi_\gamma(0)\rangle\langle\psi_\gamma(0)|,$$

and define the path-local horizontal first-order responses

$$J_{\gamma,a} = P_{\gamma,\perp}\frac{\partial}{\partial\epsilon_a}|\psi_\gamma(\epsilon)\rangle\bigg|_{\epsilon=0}, \qquad a\in\{\Omega,\Delta,V\}. \tag{7}$$

Since $\psi_\gamma(0)$ and ψ_{ref} agree only up to a global phase, align them before comparison by the common phase factor

$$c_\gamma = \frac{\langle \psi_\gamma(0)|\psi_{\mathrm{ref}}\rangle}{|\langle \psi_\gamma(0)|\psi_{\mathrm{ref}}\rangle|} \in U(1), \qquad \widetilde{J}_{\gamma,a} = c_\gamma J_{\gamma,a}, \tag{8}$$

and compare the phase-aligned response $\widetilde{J}_{\gamma,a}$ with $J_{\mathrm{ref},a}$.

Definition 1 (Numerically matched implementation fibre)**.** For a declared matching tolerance η_{match}, the numerical matched fibre $\mathcal{F}_{\eta_{\mathrm{match}}}$ consists of controls γ satisfying

$$\mathcal{I}_\gamma^{\mathrm{ref}}(0) \le \eta_{\mathrm{match}}, \qquad \|\widetilde{J}_{\gamma,a} - J_{\mathrm{ref},a}\| \le \eta_{\mathrm{match}}, \quad a \in \{\Omega, \Delta, V\}, \tag{9}$$

with $\widetilde{J}_{\gamma,a}$ the phase-aligned response of Eq. (8). The symbol η_{match} (matching tolerance) is kept distinct from the loss perturbation bound η used in Corollary 1.

3.2 Exact projective constraint map

The dynamics preserves the exchange-symmetric three-dimensional subspace. Throughout, we use the fixed normalized symmetric basis $\{|gg\rangle, (|gr\rangle+|rg\rangle)/\sqrt{2}, |rr\rangle\}$ of that subspace; in this basis we write the state as

$$\psi = (\psi_0, \psi_1, \psi_2)^T \in \mathbb{C}^3,$$

and eliminate the unobservable global phase by the projective coordinates

$$w_c = \frac{\psi_c}{\psi_0}, \qquad c = 1, 2, \tag{10}$$

valid on any region where the chart denominator ψ_0 excludes zero. For each error direction $a \in \{\Omega, \Delta, V\}$, the induced projective first response is

$$\dot{w}_{a,c} = \frac{\dot{\psi}_{a,c}\,\psi_0 - \psi_c\,\dot{\psi}_{a,0}}{\psi_0^2}, \qquad c = 1, 2, \tag{11}$$

where $\dot{\psi}_a = \partial_{\epsilon_a} \psi\big|_{\epsilon=0}$. The exact constraint map is then

$$F(\phi) = \begin{pmatrix} \Re(w - w_{\mathrm{ref}}) \\ \Im(w - w_{\mathrm{ref}}) \\ \Re(\dot{w}_\Omega - \dot{w}_{\Omega,\mathrm{ref}}) \\ \Im(\dot{w}_\Omega - \dot{w}_{\Omega,\mathrm{ref}}) \\ \Re(\dot{w}_\Delta - \dot{w}_{\Delta,\mathrm{ref}}) \\ \Im(\dot{w}_\Delta - \dot{w}_{\Delta,\mathrm{ref}}) \\ \Re(\dot{w}_V - \dot{w}_{V,\mathrm{ref}}) \\ \Im(\dot{w}_V - \dot{w}_{V,\mathrm{ref}}) \end{pmatrix} \in \mathbb{R}^{16}. \tag{12}$$

Because the chart (10) requires a nonzero denominator, the domain of F is the chart-valid open set

$$F : U \to \mathbb{R}^{16}, \qquad U = \{\gamma \in \mathbb{R}^{24} : \psi_{\gamma,0}(0) \neq 0\},$$

rather than all of $\mathbb{R}^{24}$; whenever we write $F : \mathbb{R}^{24} \to \mathbb{R}^{16}$ below it is shorthand for this chart-valid restriction. The formal accepted transverse boxes are formally verified to lie inside U (the chart denominator excludes zero on every accepted box; see Section 6). The exact matching used below is therefore exact projective output-state and first-projective-response matching: it is exact after elimination of the global phase, rather than a componentwise equality of raw state vectors. The numerical optimizer retains the full phase-aligned complex output state and the three full phase-aligned complex state tangents; redundant real components are retained in the residual vector, while singular-value decomposition determines the independent rank.

Lemma 1 (Chart validity and projective matching)**.** *Let the reference amplitude satisfy $\psi_{\mathrm{ref},0} \neq 0$ and let $\psi_\gamma(0)$ lie in the same chart, so the projective chart $w_c = \psi_c/\psi_0$ of Eq. (10) is well defined. Then*

- *(i) $w(\gamma) = w_{\mathrm{ref}}$ if and only if $[\psi_\gamma(0)] = [\psi_{\mathrm{ref}}]$ projectively;*
- *(ii) in the common chart, equality of the first projective derivatives $\dot{w}_a(\gamma) = \dot{w}_{a,\mathrm{ref}}$ for $a \in \{\Omega, \Delta, V\}$ is equivalent, after the phase alignment of Eq. (8), to equality of the horizontal first responses $\widetilde{J}_{\gamma,a} = J_{\mathrm{ref},a}$.*

Consequently a zero of the constraint map F of Eq. (12) is exactly a control matched to the reference in projective output state and complete first projective response. This conclusion concerns only the declared projective response: it makes no statement about the full Hilbert-space state vector, the unobservable global phase, or any undeclared higher-order response.

Proof. (i) Since $\psi_0 \neq 0$, the chart map $[\psi] \mapsto (\psi_1/\psi_0, \psi_2/\psi_0)$ is a well-defined bijection from the projective class onto $\mathbb{C}^2$: $w(\gamma) = w_{\mathrm{ref}}$ says $\psi_\gamma(0) = \lambda\, \psi_{\mathrm{ref}}$ with $\lambda = \psi_{\gamma,0}/\psi_{\mathrm{ref},0} \neq 0$, i.e., $[\psi_\gamma(0)] = [\psi_{\mathrm{ref}}]$, and the converse is immediate. (ii) Differentiating the quotient $w_c = \psi_c/\psi_0$ at $\epsilon = 0$ by the quotient rule gives Eq. (11). Under $\psi \mapsto c_\gamma \psi$ the numerator $\dot{\psi}_{a,c}\psi_0 - \psi_c\dot{\psi}_{a,0}$ acquires the factor c_γ^2 and the denominator ψ_0^2 the same factor, so each $\dot{w}_{a,c}$ is invariant under global phase and depends only on the horizontal component $P_{\gamma,\perp}\dot{\psi}_a$. Equality of the $\dot{w}_a$ for all a, c is therefore equality of the phase-aligned horizontal responses. Combining (i) and (ii), $F = 0$ componentwise is exactly projective output-state plus complete first-projective-response matching. □

3.3 Quadratic degeneracy under first-response matching

The next lemma is model-independent. It upgrades the empirical agreement of quadratic loss coefficients to an exact statement: projective first-response matching forces two families to share the same quadratic loss form, so their symmetrized losses can first differ at fourth order.

Lemma 2 (Quadratic degeneracy under projective first-response matching)**.** *Let $\psi_\alpha(\epsilon)$ and $\psi_\beta(\epsilon)$ be normalized analytic state families in a finite-dimensional Hilbert space such that*

$$[\psi_\alpha(0)] = [\psi_\beta(0)] = [\psi_0]$$

projectively, with identical horizontal first derivatives at the origin, $h_{\alpha,a} = h_{\beta,a}$, where

$$h_{\gamma,a} = P_\perp \frac{\partial}{\partial \epsilon_a} |\psi_\gamma(\epsilon)\rangle \bigg|_{\epsilon=0}, \qquad P_\perp = I - |\psi_0\rangle\langle\psi_0|.$$

Then

$$\big(1 - |\langle\psi_0|\psi_\alpha(\epsilon)\rangle|^2\big) - \big(1 - |\langle\psi_0|\psi_\beta(\epsilon)\rangle|^2\big) = \mathcal{O}(\|\epsilon\|^3),$$

and for the symmetrized response $S_\gamma(t, v)$ (the even part of $\mathcal{I}_\gamma^{\mathrm{loc}}$ in t, defined in Section 4),

$$S_\alpha(t, v) - S_\beta(t, v) = \mathcal{O}(t^4).$$

Proof. The infidelity is invariant under a global phase, so after phase alignment we may take $\psi_\alpha(0) = \psi_\beta(0) = \psi_0$ and use the common target ψ_0. Write $f_\gamma(\epsilon) = 1 - |\langle\psi_0|\psi_\gamma(\epsilon)\rangle|^2$ and expand

$$\langle\psi_0|\psi_\gamma(\epsilon)\rangle = 1 + \epsilon_a A_a + \tfrac{1}{2}\epsilon_a\epsilon_b A_{ab} + \mathcal{O}(\|\epsilon\|^3),$$

with $A_a = \langle\psi_0|\partial_a\psi_\gamma\rangle$ and $A_{ab} = \langle\psi_0|\partial_a\partial_b\psi_\gamma\rangle$ evaluated at $\epsilon = 0$; repeated indices are summed. Differentiating the normalization $\langle\psi_\gamma|\psi_\gamma\rangle = 1$ gives $\Re A_a = 0$ and $\Re A_{ab} = -\Re\langle\partial_a\psi_\gamma|\partial_b\psi_\gamma\rangle$. The linear term of f_γ therefore vanishes, and its quadratic term is the Fubini–Study form of the horizontal derivatives,

$$f_\gamma(\epsilon) = g_{ab}\,\epsilon_a\epsilon_b + \mathcal{O}(\|\epsilon\|^3), \qquad g_{ab} = \Re\langle h_{\gamma,a}|h_{\gamma,b}\rangle,$$

since $h_{\gamma,a} = \partial_a\psi_\gamma - A_a\psi_0$. Identical horizontal first derivatives give $g_{ab}^\alpha = g_{ab}^\beta$, hence $f_\alpha - f_\beta = \mathcal{O}(\|\epsilon\|^3)$. Symmetrization in t cancels the odd-order terms, so the difference of the symmetric responses starts at order t^4. □

3.4 Dimension count and generation

The control has 24 phase variables. At the reference solution, the numerical constraint Jacobian has rank

$$\operatorname{rank} DF(\gamma_{\mathrm{ref}}) = 16,$$

leaving an eight-dimensional numerical null space that organizes the candidate implementation fibre. Candidate paths are generated by:

1. drawing a seeded direction in the numerical null space;
2. applying a finite displacement;
3. projecting back to the constraint set by nonlinear least squares;
4. rejecting candidates whose constraint residual or pairwise phase distance fails a predeclared gate.

No held-out finite-error outcome is evaluated during candidate generation. The maximum constraint residual in the reported runs is of order 10^{-11}.

Remark 1 (Global fibre versus local certified roots). The numerical rank and residual establish a highly accurate computational fibre, suggesting a locally eight-dimensional numerical implementation fibre, not the exact existence of a global eight-dimensional solution manifold. A complete global fibre theorem would require a square transverse formulation and a global interval-Newton or Krawczyk proof. The present work instead establishes 12 locally unique roots exactly matched in projective output state and first projective response via strict Krawczyk inclusions in transverse boxes. The global fibre structure remains open.

Figure 1 summarizes the local matching and quartic-separation mechanism.

4 Prospective fourth-order response prediction

4.1 Symmetric local response

Introduce normalized error coordinates $z \in \mathbb{R}^3$ through

$$\epsilon = Dz, \qquad D = \operatorname{diag}(0.06, 0.04, 0.05).$$

For a direction $v \in \mathbb{R}^3$, define the symmetric response

$$S_\gamma(t, v) = \frac{\mathcal{I}_\gamma^{\mathrm{loc}}(tDv) + \mathcal{I}_\gamma^{\mathrm{loc}}(-tDv)}{2}. \tag{13}$$

Analyticity of the finite matrix exponential gives

$$S_\gamma(t, v) = q_{2,\gamma}(v)t^2 + q_{4,\gamma}(v)t^4 + q_{6,\gamma}(v)t^6 + \cdots . \tag{14}$$

Odd powers cancel by construction.

By Lemma 2, exact projective first-response matching forces the quadratic loss forms of matched controls to coincide exactly, so the quartic term is the first possible symmetric discriminator. The numerical cohorts here are matched only within tolerance; consistently, the relative spread of the averaged quadratic coefficient across the validation cohorts is approximately 10^{-7}. The first useful discriminator is therefore the quartic coefficient.

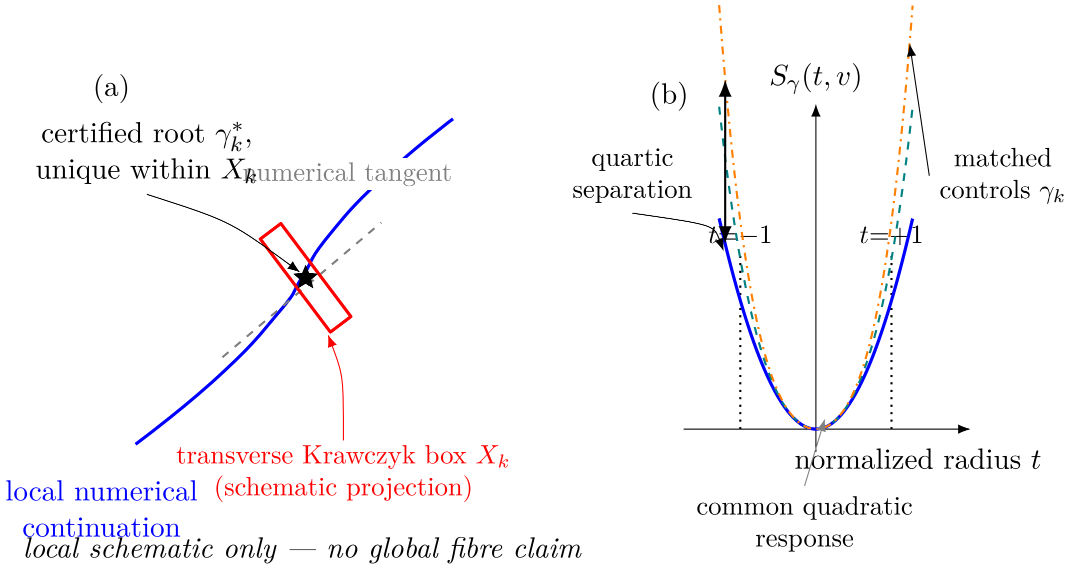

Figure 1: Schematic of local response matching and quartic separation (illustrative; not a proof figure). (a) In a local control-space chart, the projective output-state and first-response constraints define a numerical tangent to the numerical continuation; the red rectangle is a schematic projection of the transverse Krawczyk box X_k, drawn crossing the tangent transversally, inside which a locally unique certified root γ_k^* is established. The panel makes no global connected-fibre claim. (b) Quartic mechanism schematic. Exact projective first-response matching makes the quadratic loss form common across matched controls γ_k, so symmetrized finite-error responses first separate through their quartic coefficients between the declared normalized probe boundaries $t = \pm 1$. This panel illustrates the low-order discrimination mechanism only; it is not the direct outward-rounded Arb propagation that proves the main ordering theorem (Theorem 2). The tensor $\mathcal{A}^{(4)}$ is a fourth-order response susceptibility, not a Riemannian curvature tensor.

4.2 Prospective quartic ranking

The quartic rule developed here is a *geometric predictor*, not a universal theorem: it is gap-dependent and certifies only a subset of pairwise comparisons, a boundary made precise in Sections 5–6.

4.2.1 Discovery and validation separation

The initial four named controls were used only for discovery and orientation. A twelve-path cohort was then used to compare candidate formulas. The smallest sufficient rule on that training cohort was

$$\text{smaller } \mathcal{G}_4 \implies \text{smaller mean finite-error infidelity.}$$

The addition of the sixth-order coefficient did not improve the training rank correlation, so the quartic rule was frozen.

For the prospective audit, the coefficients q_2, q_4, q_6 were estimated by a frozen least-squares fit of

$$S_\gamma(t, e_a) = q_{2,a}t^2 + q_{4,a}t^4 + q_{6,a}t^6$$

at

$$t \in \left\{\tfrac{1}{32}, \tfrac{1}{24}, \tfrac{1}{20}, \tfrac{1}{16}, \tfrac{1}{14}, \tfrac{1}{12}, \tfrac{1}{10}, \tfrac{1}{8}\right\}.$$

The tensor audit below is computationally distinct: its directional derivatives are evaluated directly at $t = 0$ by block-matrix exponentials, rather than by fitting nonzero-radius samples.

A new twenty-path cohort was generated with a different seed. For each candidate, a ranking certificate containing phases, local coefficients, and predicted order was written and SHA-256 hashed before the held-out outcome evaluator was enabled. The predeclared primary gate was

$$\rho_{\text{Spearman}} \geq 0.95,$$

together with top-path agreement within each recorded execution.

4.2.2 Quartic validation result

The prospective validation result is therefore reported at the predeclared threshold level:

$$\rho_{\text{Spearman}}\big(-\mathcal{G}_4, -\bar{\mathcal{I}}\big) \geq 0.95. \tag{15}$$

Across the recorded architecture-specific executions ($n = 2$) of the prospective cohort-generation-and-evaluation protocol, the predeclared rank gate was satisfied. The arm64 reference environment recorded $\rho = 0.99398$ with top path pv17, and the x86_64/Rosetta diagnostic environment recorded $\rho = 0.98947$ with top path pv02; thus the exact correlation, complete ranking, and named top path were not architecture-invariant, while both environments cleared the $\rho \geq 0.95$ gate. This empirical evidence is not used in the proof of the exact-root ordering theorem and does not establish a universal fourth-order ranking law. Because the quartic rule, ranking direction, acceptance threshold, and ranking certificate were frozen before held-out finite-error evaluation, the threshold-level result is not an in-sample fit; nevertheless, it remains cohort-specific prospective evidence rather than a theorem. Two local coefficient-extraction windows had rank correlation one, and the maximum relative fit residual was 2.37×10^{-10}, the direction-safe upward rendering of the frozen field `maximum_relative_fit_residual` in `results/g4_prospective/report.json`.

Subsequent cohorts show that $\mathcal{G}_4$ is not universally sufficient: depending on the generated matched paths, the prospective quartic correlation varies appreciably across seeded cohorts. This variation motivated the bounded higher-order analysis rather than post hoc replacement of the quartic formula.

5 Covariant quartic response geometry

5.1 Quartic tensor

The object recovered in this section is a quartic susceptibility of the endpoint loss—a component of the fourth jet of the endpoint map composed with infidelity—not a Riemannian curvature or an instance of the quantum geometric tensor [19]. There exists a symmetric fourth-order tensor $\mathcal{A}^{(4)}_{\gamma}$ such that

$$q_{4,\gamma}(v) = \mathcal{A}^{(4)}_{\gamma,ijkl}\, v_i v_j v_k v_l. \tag{16}$$

In three dimensions a symmetric fourth-order tensor has $\binom{3+4-1}{4} = 15$ independent components. The tensor $\mathcal{A}^{(4)}_{\gamma}$ is reconstructed from the fourth jet of the path-local loss $\mathcal{I}^{\text{loc}}_{\gamma}$: we recover its components from 36 seeded directions, whose directional Taylor coefficients are computed directly at $t = 0$ through order six by block-matrix exponentials, rather than by fitting nonzero-radius samples.

Let

$$z \sim \text{Unif}\{\pm e_{\Omega}, \pm e_{\Delta}, \pm e_V\}, \qquad \epsilon = Dz,$$

so that the physical error scales $0.06^4, 0.04^4, 0.05^4$ are carried by D and absorbed into the tensor components. For the six-axis law in Eq. (5), define the fourth moment

$$M^{(4)}_{ijkl} = \mathbb{E}[z_i z_j z_k z_l] \,.$$

The scalar quartic predictor is

$$\mathcal{G}_4(\gamma; M^{(4)}) = \mathcal{A}^{(4)}_{\gamma,ijkl} M^{(4)}_{ijkl}. \tag{17}$$

For the equally weighted signed axes,

$$\mathcal{G}_4 = \frac{1}{3}\Big(\mathcal{A}^{(4)}_{\Omega\Omega\Omega\Omega} + \mathcal{A}^{(4)}_{\Delta\Delta\Delta\Delta} + \mathcal{A}^{(4)}_{VVVV}\Big).$$

5.2 Coordinate covariance

Proposition 1 (Covariance and invariant contraction)**.** *Let $z = Ry$ for an invertible real matrix R. Then*

$$\mathcal{A}^{(4)}_{y,abcd} = \mathcal{A}^{(4)}_{z,ijkl}\, R_{ia} R_{jb} R_{kc} R_{ld}, \tag{18}$$

$$M^{(4)}_{y,abcd} = (R^{-1})_{ai}(R^{-1})_{bj}(R^{-1})_{ck}(R^{-1})_{dl}\, M^{(4)}_{z,ijkl}, \tag{19}$$

and

$$\mathcal{A}^{(4)}_y : M^{(4)}_y = \mathcal{A}^{(4)}_z : M^{(4)}_z. \tag{20}$$

Proof. Equation (18) follows by substituting $z = Ry$ into the quartic form. Equation (19) follows from $y = R^{-1}z$. Contracting all four indices cancels each R with the corresponding R^{-1}, proving Eq. (20). □

This proposition is algebraic. The numerical audit additionally recomputes the tensor directly in the transformed coordinates. For a nonorthogonal map of condition number approximately 1.82, the direct-refit relative error is 1.08×10^{-14} and the maximum invariant-contraction relative error is 2.36×10^{-15}.

The covariant object is not the scalar $\mathcal{G}_4$ in isolation. It is the fourth-order response tensor $\mathcal{A}^{(4)}$ together with the physical fourth noise moment $M^{(4)}$. Their contraction is independent of a linear reparameterization of error coordinates. This avoids interpreting a coordinate-specific axis average as an intrinsic curvature.

Establishing a relation of this susceptibility to a canonical connection or curvature on a control quotient would require additional structure.

For a noise moment M, the low-cost screening problem is

$$\gamma^\star \in \arg\min_{\gamma\in\mathcal{F}_{\eta_{\mathrm{match}}}} \ \mathcal{A}^{(4)}_\gamma : M^{(4)}.$$

Pairs that fail the quartic gap test can be escalated adaptively to sixth, eighth, or higher jet order until their intervals separate. This suggests a certified branch-and-bound control selector: compute low-order local tensors for all candidates, refine only ambiguous pairs, and reserve direct finite-error simulation or hardware evaluation for model validation rather than exhaustive ranking.

The gap-dependent certification criterion used by the implementation retains the residual constant and quadratic terms explicitly. The certified quantity is the six-axis averaged reference loss at radius t,

$$\mathcal{I}^{\mathrm{ref}}_k(t) = \frac{1}{6}\sum_{a\in\{\Omega,\Delta,V\}}\sum_{\sigma=\pm1} \mathcal{I}^{\mathrm{ref}}_k(\sigma t D e_a). \tag{21}$$

At unit radius $t = 1$ the six signed probe points σDe_a are exactly the six signed axial errors of the declared error set, so Eq. (21) reduces to the six-axis mean functional of Eq. (6); the radius-t form is its one-parameter rescaling used for the local jet analysis. For each path k write the truncated local model and its enclosure as

$$P_k(t) = C_{0,k} + C_{2,k}t^2 + \mathcal{G}_{4,k}t^4, \qquad \mathcal{I}_k^{\mathrm{ref}}(t) \in P_k(t) + [-B_k(t),\, B_k(t)]. \tag{22}$$

If

$$|P_a(t) - P_b(t)| > B_a(t) + B_b(t), \tag{23}$$

then the sign of $\mathcal{I}_a^{\mathrm{ref}}(t) - \mathcal{I}_b^{\mathrm{ref}}(t)$ is strictly certified. Only when $C_{0,a} = C_{0,b}$ and $C_{2,a} = C_{2,b}$ does this reduce to the simplified quartic-gap form

$$|\mathcal{G}_{4,a} - \mathcal{G}_{4,b}|\, |t|^4 > B_a(t) + B_b(t).$$

The implementation folds the residual C_0 and C_2 differences into the correction intervals, so they are not omitted from the certificate.

6 Computer-assisted certification

6.1 Krawczyk root theorem

The numerical paths satisfy the endpoint and first-order matching constraints to order 10^{-11} (see Remark 2 for the precise norm and its relation to the formal boxes). To move from numerical candidates to exact mathematical roots, we use the projective constraint map $F : \mathbb{R}^{24} \to \mathbb{R}^{16}$ of Eq. (12). We do *not* split the 24 original phase coordinates into a fixed subset of 16 transverse coordinates and 8 retained fibre coordinates. Instead, for each frozen path k the certificate uses the affine transverse parameterization

$$\gamma = \widehat{\gamma}_k + N_k\, x, \qquad x \in X = [-3 \times 10^{-12},\, 3 \times 10^{-12}]^{16} \subset \mathbb{R}^{16}, \tag{24}$$

where $\widehat{\gamma}_k \in \mathbb{R}^{24}$ is the frozen box centre of path k and the columns of the frozen 24×16 matrix N_k form the declared orthonormal transverse basis. The variable x is the vector of transverse-basis coefficients, not a selection of 16 of the original 24 phase coordinates, and N_k varies from path to path. On this declared affine transverse slice we consider the reduced square map

$$F_k(x) = F(\widehat{\gamma}_k + N_k x), \qquad F_k : \mathbb{R}^{16} \to \mathbb{R}^{16},$$

whose Jacobian is the reduced 16×16 matrix

$$DF_k(x) = DF(\widehat{\gamma}_k + N_k x)\, N_k \in \mathbb{R}^{16\times 16}. \tag{25}$$

All interval operators below act on this path-dependent reduced map; at no point is a transverse vector $x \in \mathbb{R}^{16}$ passed directly to $F : \mathbb{R}^{24} \to \mathbb{R}^{16}$. For a box $X \subset \mathbb{R}^{16}$ with centre x_0, the Krawczyk operator for path k is

$$K_k(X) = x_0 - C_k F_k(x_0) + \big(I - C_k[DF_k(X)]\big)(X - x_0), \tag{26}$$

where C_k is the frozen point preconditioner of the corresponding reduced midpoint Jacobian $DF_k(x_0)$ (a numerical approximate inverse) and $[DF_k(X)]$ is an interval enclosure of the reduced Jacobian (25) over X. In centered coordinates $x_0 = 0$, as used by the implementation, this reduces to

$$K_k(X) = -C_k F_k(0) + \big(I - C_k[DF_k(X)]\big)X.$$

The strict inclusion $K_k(X) \subset \mathrm{int}(X)$ then implies that F_k has a unique zero inside X [20, 9]. The projective chart denominator ψ_0 of Eq. (10) is formally verified to exclude zero on every accepted box, ensuring the coordinate chart is valid throughout the Krawczyk computation.

Theorem 1 (Krawczyk inclusions for 12 declared controls)**.** *For each of the twelve declared controls, with the affine transverse parameterization of Eq.* (24)*, the common transverse box*

$$X_k = X = [-3 \times 10^{-12}, 3 \times 10^{-12}]^{16}$$

satisfies the strict Krawczyk inclusion

$$K_k(X_k) \subset \operatorname{int}(X_k).$$

Consequently, each box X_k contains a unique exact root x_k^* of $F_k(x) = 0$, i.e., a control $\gamma_k^* = \widehat{\gamma}_k + N_k x_k^*$ that is exactly matched in projective output state and first projective response to the reference. Existence and uniqueness are certified within the declared transverse box only, not on the full 24-dimensional phase space or the whole response fibre.

Computer-assisted proof. A numerical approximate inverse of the midpoint Jacobian is used as the point preconditioner. The Jacobian over each full box is then enclosed with 192-bit Arb arithmetic, and the resulting Krawczyk image is verified to lie strictly inside the box. The projective chart denominator is formally verified to exclude zero on every accepted box. For all twelve boxes, the inclusion $K_k(X) \subset \operatorname{int}(X)$ is verified; the worst box-normalized utilization, read from the frozen certificate, is

$$\max_k \frac{\|K_k(X)\|_\infty}{3 \times 10^{-12}} < 0.866 < 1,$$

so every coordinate image clears the box wall by a factor of at least 1.15. The SHA-256 protocol and certificate identifiers of this run are recorded in Appendix B. By the Krawczyk theorem [20, 9, 10], each box contains a locally unique root. As a *supplementary* audit-closure check—not an additional premise of the strict inclusion just established—the regularity of each frozen point preconditioner is independently certified by a 256-bit outward-rounded Rump–Neumann argument: for all twelve paths $\|I - R_k C_k\|_\infty < 1$, so each C_k is nonsingular (certificate `results/audit_closure/p0_preconditioner_certificate.json`). This P0 certificate belongs to the later v0.3.2 audit-closure supplement (Remark 5); it strengthens auditability but neither replaces nor alters the v0.3.1 frozen Krawczyk certificate that proves the theorem. In the strict-inclusion form of the Krawczyk theorem used here, the required regularity is already implied by the accepted inclusion; the v0.3.2 P0 certificate is therefore an independent redundant audit of the production preconditioner, not an additional premise of Theorem 1. □

Remark 2 (Residual, norm, and box relation). Two distinct residuals must not be conflated. The *unpreconditioned* constraint residual $\|F\|$ of the numerical candidates is of order 10^{-11}; it is a float64 diagnostic of the candidate-generation optimizer and is *not* compared directly with the 3×10^{-12} box radius. The strict Krawczyk inclusion is decided instead by the full interval image $-C_k F_k(x_0) + \big(I - C_k\,[DF_k(X)]\big)(X - x_0)$ of Eq. (26), whose outward-rounded endpoints are enclosed on the formal cohort box centres $\widehat{\gamma}_k$. The preconditioned image $\|K_k(X)\|_\infty$, already preconditioned by C_k, is what the utilization bound 0.866 above reports. The frozen certificate does not tabulate a single scalar preconditioned centre residual; accordingly we do not quote one, and the 10^{-11} figure refers only to the unpreconditioned optimizer residual.

Remark 3 (Scope of the Krawczyk inclusion theorem). The theorem establishes local uniqueness inside each declared transverse box. It does not certify global uniqueness on the entire fibre, nor does it prove that every point on the numerical fibre has an exact representative. The global fibre structure remains open.

6.2 Direct finite-error ordering theorem

The following theorem is the main result of this work. Its proof uses only the certified Krawczyk boxes and direct outward-rounded propagation of the finite-error functional. No Taylor truncation, Cauchy extraction, alias enclosure, or local-jet reconstruction is used.

Theorem 2 (Exact-root finite-error ordering)**.** *For each of the twelve declared controls, let X_k be the transverse phase box satisfying a strict Krawczyk inclusion from Theorem 1. Direct outward-rounded Arb propagation of X_k through the declared six-point finite-error functional produces twelve performance intervals B_k covering the complete root box,*

$$B_k \supseteq \left\{ \overline{\mathcal{E}}\left(\widehat{\gamma}_k + N_k x\right) : x \in [-3 \times 10^{-12},\, 3 \times 10^{-12}]^{16} \right\},$$

that are pairwise disjoint and totally ordered according to the pre-outcome ordering frozen for the independent twelve-path formal cohort. Therefore the locally unique roots exactly matched in projective output state and first projective response inside the boxes obey the complete frozen finite-error ordering.

Computer-assisted proof. By Theorem 1, the locally unique projectively matched control $\gamma_k^* = \widehat{\gamma}_k + N_k x_k^*$ belongs to the certified box. For every box X_k, the six declared finite-error Hamiltonians are evaluated directly using 192-bit outward-rounded Arb/ACB arithmetic over the *whole* declared transverse box, not merely at its centre or at the approximate root: every phase-dependent trigonometric factor, Hamiltonian matrix exponential, state propagation, overlap, infidelity, and six-point mean is enclosed. Denote the resulting real performance interval by B_k. Because the propagation is over the full box,

$$B_k \supseteq \left\{ \overline{\mathcal{E}}\left(\widehat{\gamma}_k + N_k x\right) : x \in [-3 \times 10^{-12},\, 3 \times 10^{-12}]^{16} \right\},$$

where $\overline{\mathcal{E}}$ is the six-axis mean finite-error functional of Eq. (6). Since the certified root satisfies $x_k^* \in X$, its exact-root performance lies in the enclosure,

$$\bar{\mathcal{I}}(\gamma_k^*) \in B_k.$$

No Taylor truncation, order-30 jet, or local-jet reconstruction enters B_k. For every frozen ordered pair $a \prec b$, direct interval comparison verifies

$$\sup B_a < \inf B_b.$$

All $\binom{12}{2} = 66$ comparisons pass and have the orientation specified by the frozen ordering certificate. Because each B_k encloses its exact root's performance, the strict box separation $\sup B_a < \inf B_b$ implies the strict exact-root ordering

$$\bar{\mathcal{I}}(\gamma_a^*) < \bar{\mathcal{I}}(\gamma_b^*)$$

for every frozen ordered pair. □

The predecessor order was frozen before direct propagation; the resulting protocol and certificate hashes are reported in Appendix B.

Remark 4 (Scope of Theorem 2)*.* The theorem is conditional on the serialized-decimal finite Hamiltonian model, the declared transverse boxes, and the six-axis mean error functional. It does not certify model discrepancy, global fibre topology, many-body scaling, or hardware. It also does not prove a global eight-dimensional exact fibre theorem; it proves finite-error ordering for the locally unique exact roots inside the declared boxes.

Corollary 1 (Ordering stability under bounded loss discrepancy). *Let B_k denote the certified direct performance interval for the kth exact root, and define the minimum certified ordering margin*

$$\Delta_{\min} = \min_{a \prec b} \bigl(\inf B_b - \sup B_a\bigr) > 0,$$

where $a \prec b$ ranges over the frozen ordered path pairs. Suppose an additional physical effect changes the evaluated loss of every fixed certified control by at most η:

$$\bigl|\tilde{\bar{\mathcal{I}}}(\gamma_k^*) - \bar{\mathcal{I}}(\gamma_k^*)\bigr| \le \eta \quad \textit{for every } k.$$

If

$$2\eta < \Delta_{\min},$$

then the complete frozen ordering of the twelve fixed certified controls is preserved under the perturbed loss.

Proof. For every frozen ordered pair $a \prec b$,

$$\tilde{\bar{\mathcal{I}}}(\gamma_a^*) \le \sup B_a + \eta < \inf B_b - \eta \le \tilde{\bar{\mathcal{I}}}(\gamma_b^*),$$

where the strict middle inequality follows from $2\eta < \Delta_{\min}$. □

For the frozen twelve-root certificate, the minimum formally certified pairwise margin, computed from the outward-rounded endpoints of the frozen direct-certificate intervals, is

$$\Delta_{\min} = 2.50 \times 10^{-5},$$

attained by the pair pv08 and pv11. For the certified minimum separation $\Delta_{\min} = 2.505874\ldots \times 10^{-5}$, the sufficient uniform perturbation condition in Corollary 1 becomes $\eta < \Delta_{\min}/2 \approx 1.25 \times 10^{-5}$. This is a stringent model-discrepancy requirement and should not be interpreted as a certificate of hardware-level ordering robustness.

Corollary 1 is a conditional stability statement for the losses evaluated at the same certified controls. It does not prove that an added Lindblad term, waveform filter, calibration drift, or other model perturbation satisfies the required bound, nor that the controls remain exact response-matched roots of the perturbed model. Proving root continuation under model perturbation would require a parameterized Krawczyk or validated implicit-function argument.

6.3 Order-thirty mechanism certificate

The following proposition provides a separate, computationally distinct mechanism certificate using the zero-error jet. Unlike the main theorem, it uses Taylor coefficients, Cauchy extraction, alias enclosures, and Dyson tail bounds. It is not part of the main theorem proof.

The floating-point precursor obtains the order-thirty jet by block-matrix exponentials. The formal phase-box certificate used in Proposition 2 instead extracts the coefficients with a 64-point outward-rounded Cauchy transform.

Proposition 2 (Order-thirty mechanism certificate). *Propagation of the zero-error jet through order thirty over the twelve Krawczyk phase boxes, including the formal Cauchy-alias and analytic order-32 tail enclosures, certifies 52/66 frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified.*

Computer-assisted proof. For path k and physical error axis a, define the analytic transition amplitude

$$g_{k,a}(\lambda) = \langle \psi_{\mathrm{ref}} | \psi_{\gamma_k}(\lambda D e_a) \rangle = \sum_{n \geq 0} g_{k,a,n} \, \lambda^n.$$

Its coefficients are extracted by a 64-point outward-rounded Cauchy discrete Fourier transform on the complex unit circle:

$$\widehat{g}_n = \frac{1}{64} \sum_{k=0}^{63} g(\zeta_k) \, \zeta_k^{-n}, \qquad \zeta_k = e^{2\pi i k/64}. \tag{27}$$

The discrete transform aliases coefficients $g_{n+64\ell}$. A path-uniform Dyson bound [8] encloses this alias. The implemented enclosure is 10^{-40}, and the formal gate verifies that

$$\frac{e^{K_a} K_a^{64}}{64!} < 10^{-40}$$

for every physical error axis, where the dependence is parameterized as

$$H_a(s; \lambda_a) = H_0(s) + \lambda_a H_{1,a}(s),$$

with s the physical time and λ_a the dimensionless error parameter along axis a, and

$$K_a = \int_0^T \| H_{1,a}(s) \|_2 \, ds = \int_0^T \left\| \left. \frac{\partial H_a(s; \lambda_a)}{\partial \lambda_a} \right|_{\lambda_a = 0} \right\|_2 ds$$

is the integrated spectral norm of the Hamiltonian perturbation along axis a. Fidelity coefficients are formed by ball convolution,

$$F_n = \sum_{r=0}^{n} g_r \, \bar{g}_{n-r},$$

and the infidelity coefficients follow from $1 - F$. The mean six-axis analytic even Dyson tail after order thirty is enclosed by 2×10^{-11}; the largest individual-axis tail is below 4×10^{-11}.

Propagating the resulting order-thirty jet over the twelve Krawczyk phase boxes yields performance intervals. Pairwise comparison of the resulting order-thirty intervals certifies 52/66 frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified. The unresolved pairs arise from dependency and wrapping in the phase-box propagation of the jet enclosure. They are not reversed physical comparisons and do not indicate an omitted-tail failure. □

The three rows reporting pair counts are not interchangeable. The quartic-only 34/66 and the frozen-point order-30 66/66 are *point-valued* results at the serialized frozen decimal points; the exact-root order-30 52/66 and the direct 66/66 are *box-valued* certificates over the certified Krawczyk root boxes, and only the last is the main theorem. The covariant $\rho = 0.965035$ entry is a reference-environment numerical value on the twelve-path covariance/ranking audit; it is a separate diagnostic, not a second application of the predeclared prospective gate. The predeclared $\rho \geq 0.95$ gate applies to the independent twenty-path prospective cohort and is satisfied.

Table 1: Logical hierarchy of the prospective and formal audits. "Rank diagnostic" denotes ranking correlation, while "Certified result" identifies the formally enclosed object ("cert." = formally certified pairs; "numerical" = point-valued diagnostic). Cohorts differ across stages and all protocols are frozen before held-out evaluation. The final row is the main theorem; covariance residuals are reported in Appendix B.

Audit	Cohort	Rank diagnostic	Certified result	Logical role
Prospective $\mathcal{G}_4$ ranking	20-path	$\rho \geq 0.95$ gate	–	prospective evidence
Covariant $\mathcal{G}_4$ contraction ranking	12-path	$\rho = 0.965035$	–	covariance/ranking audit
Formal quartic-only bound	12-path	–	34/66	low-order boundary (point)
Frozen point-valued order-30 diagnostic	12-path	$\rho = 1$	66/66 numerical	point-valued diagnostic
Exact-root Krawczyk inclusion	12-path	–	12/12 roots	existence/uniqueness
Exact-root order-30 jet	12-path	–	52/66 cert.	mechanism certificate (box)
Exact-root direct Arb propagation	12-path	$\rho = 1$	**66/66 cert.**	**main theorem (box)**

7 Summary of certified and prospective results

Table 1 separates the main theorem, its local-root prerequisite, the higher-order mechanism certificate, and the independent prospective evidence; Figure 2 gives the exact-root data-level statement, while the prospective ranking result is reported only at threshold level in the text. One distinction must be made explicit: the frozen-point order-30 audit resolves 66/66 pair directions at the frozen decimal points, before any propagation over the Krawczyk boxes, and is a point-valued numerical diagnostic, not an exact-root box certificate (proof objects in `results/l4_formal/`, together with the quartic-only 34/66 boundary); propagation of the same jet over the root boxes resolves 52/66 comparisons with no reversal (Proposition 2, box-valued), whereas direct finite-error propagation over those boxes proves all 66/66 (box-valued, the main theorem). Numerical integrity diagnostics are collected in Appendix B. Table 2 maps each claim layer to its cohort, mathematical object, and artifact path.

Table 2: Claim-layer to artifact mapping. "Point" = point-valued diagnostic at frozen decimal points; "box" = box-valued interval certificate over the certified root boxes. The training, formal twelve-path, and prospective twenty-path cohorts are distinct frozen sets with distinct seeds.

Claim/layer	Cohort	Object	Artifact path	Value type
Quartic-only 34/66	12-path formal	quartic gap	`results/l4_formal/`	point
Frozen order-30 66/66	12-path formal	order-30 point	`results/l4_formal/`	point
Exact-root order-30 52/66	12-path formal	jet over boxes	`results/exact_root_ordering/`	box
Krawczyk 12/12 roots	12-path formal	strict inclusion	`results/exact_fibre_krawczyk/`	box
Direct exact-root 66/66	12-path formal	box propagation	`results/exact_root_ordering/`	box
Prospective $\rho \geq 0.95$	20-path prospective	rank gate	`results/g4_prospective/`	rank
Training (formula selection)	12-path training	rule freeze	`results/g4_prospective/`	rank
Covariance contraction ranking	12-path covariance	contraction rank	`results/l3_covariance/`	rank
P0 preconditioner audit	v0.3.2 supplement	redundant regularity	`results/audit_closure/`	audit

The covariance contraction ranking row is the 12-path covariance audit of Table 1 (recorded $\rho = 0.965035$); it is a rank diagnostic, not a gate application. The P0 row is the independent redundant regularity audit of the production preconditioner belonging to the v0.3.2 audit-closure supplement (Remark 5); it is not a new premise of the main theorem.

The twelve-path *formal* cohort frozen for the Krawczyk and ordering certificates is a decimal-refrozen cohort whose phase values differ from the earlier float64 numerical training/validation cohort by less than 6.06×10^{-7} per coordinate ($\max|\Delta\gamma| < 6.06 \times 10^{-7}$; the recorded maximum coordinate mismatch is $6.0520531 \ldots \times 10^{-7}$, so 6.06×10^{-7} is a direction-safe upward quotation). This cohort drift is far below every certified interval half-width and every nonzero ordering

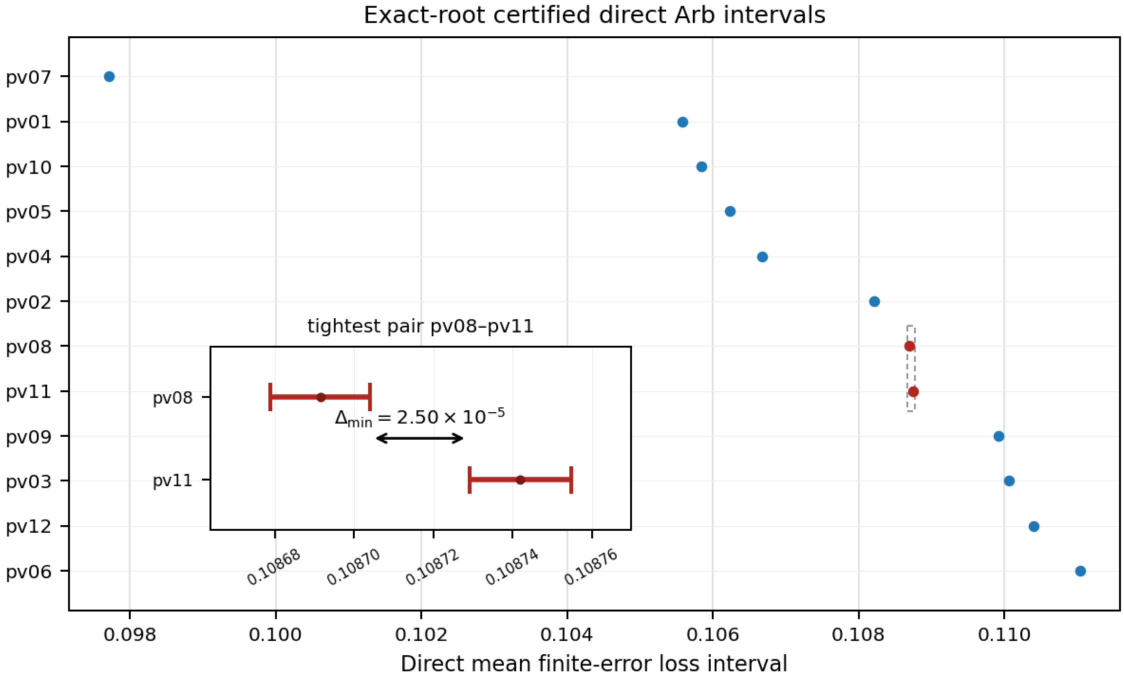

Figure 2: Certificate-derived visualization of the exact-root ordering. The twelve certified direct performance intervals B_k from 192-bit outward-rounded Arb propagation are drawn to scale in the frozen order; the prospective $\mathcal{G}_4$ result is reported only at threshold level in the text. All 66 unordered pairs are formally disjoint. In the global plot the certified interval widths are near the pixel scale, so each B_k renders as an essentially point-like marker; the inset therefore zooms the boxed region to resolve the tightest gap. The inset shows the two highlighted intervals of the minimum-gap pair pv08–pv11 with their endpoint caps and midpoints, and a double-headed arrow connecting $\sup B_{\text{pv08}}$ to $\inf B_{\text{pv11}}$; this separation is the certified margin $\Delta_{\min} = 2.50 \times 10^{-5}$ of Corollary 1. All intervals, the minimum-gap pair, and $\Delta_{\min}$ are derived from the frozen exact-root ordering certificate.

margin in the formal certificate (whose smallest margin is $\Delta_{\min} = 2.50 \times 10^{-5}$), so it does not alter any certified interval inequality or theorem-bearing certificate: the frozen theorem certificate is unchanged by it. The formal cohort is identified by the cohort hash of Appendix B; the prospective twenty-path cohort and the twelve-path training cohort are separately seeded and are not the same set.

8 Discussion

8.1 Why the quartic term is useful but incomplete

First-order matching makes the quadratic loss nearly common—exactly common at exact matched roots by Lemma 2. This promotes the quartic term to the first visible discriminator. It explains why $\mathcal{G}_4$ often gives almost perfect rankings. However, if two quartic coefficients are close, the sixth and higher terms can reverse their order. The formal intervals expose this distinction directly: 34 pairs are quartic-certified, while the remaining pairs require higher jet data or direct propagation.

This is preferable to declaring the quartic rule either universally true or false. The correct statement is gap-dependent as in Eq. (23).

8.2 Minimal model versus scalable certification

The two-atom model isolates response matching in the smallest interacting Rydberg system and establishes a minimal rigorous existence-and-ordering result, with residual finite-error ordering not attributable to optimizer residuals. Many-body extension requires a separate scalable representation and validated propagation architecture, as quantified in Section 9.

8.3 Order-30 mechanism versus direct propagation

The order-30 jet and the direct propagation answer the same comparison at different levels. The gap between their certified counts is methodological rather than contradictory: Taylor-coefficient extraction accumulates interval widening in the phase-box propagation of the jet enclosure, whereas direct propagation encloses the finite-radius functional without reconstructing its local jet. The 14 unresolved pairs are therefore not reversed in the declared model, and no omitted-tail failure is involved; the 52/66 mechanism certificate and the 66/66 direct certificate should not be conflated, the latter being the main theorem.

The particular ordering of the twelve serialized controls is not presented as a universal physical law. The reusable contribution is the certificate architecture: transverse exact-root isolation, formal propagation of the resulting root boxes, and explicit separation between low-order mechanism certificates and direct finite-error theorems.

The certification architecture is not specific to the present pulse family: it applies whenever a finite-dimensional differentiable response map admits a validated full-rank Jacobian enclosure, local root boxes can be certified, and the target observable can be propagated with outward rounding. The present work establishes this chain for one frozen two-atom model only; it makes no scaling or multi-model universality claim.

9 Limitations and next theorem

The present work has five explicit limitations.

1. **Global fibre structure.** Twelve locally unique roots exactly matched in projective output state and first projective response are certified in transverse boxes. The global eight-dimensional solution manifold structure remains unproved.
2. **Finite-dimensional closed model.** Spontaneous emission, Doppler effects, leakage, waveform filtering, and other model-discrepancy terms are absent. An open-system extension requires a new density-matrix or channel constraint map (whose dimension generally scales with d^2 rather than d), new matched controls and root boxes, and validated Liouvillian propagation; the present ordering need not survive arbitrary dissipative perturbations, and no noise-model-independent ordering theorem is claimed.
3. **Two atoms.** No many-body scaling theorem is claimed.
4. **Mean axial error law.** The primary result concerns the declared six-axis mean. Worst-case ranking is not supported by the present scalar.
5. **No QPU evidence.** The calculations are local and require no hardware access. They do not constitute physical-QPU evidence.

9.1 Constraint dimension and scalability

The present dense interval implementation is not a scalable many-body compiler. Its dimensional growth can nevertheless be stated explicitly. Suppose the reachable complex state space has dimension d, and let m be the number of calibrated error coordinates. Matching one projective output state and its m complete projective first responses imposes

$$q = 2(m + 1)(d - 1)$$

real projective constraints. When the constraint Jacobian has rank q, a square Krawczyk chart uses q transverse control variables, while any remaining independent control variables parameterize numerical fibre directions.

Here $m = 3$ and the exchange-symmetric two-atom state space has $d = 3$, giving

$$q = 2(3 + 1)(3 - 1) = 16,$$

with $24 - 16 = 8$ numerical fibre directions. For a fully permutation-symmetric N-atom two-level model, $d = N + 1$ and the corresponding constraint count is $8N$. For a generic unsymmetrized model, $d = 2^N$ and the dense constraint count becomes $8(2^N - 1)$.

Thus exponential growth is not intrinsic to the Krawczyk operator alone; it arises through the dimension of the reachable state representation and the cost of enclosing the associated propagators and Jacobians. In generic many-body settings, direct dense Arb/ACB propagation and naive interval Jacobians would also suffer rapidly increasing dependency and wrapping. Possible extensions include symmetry-reduced state representations, structured validated propagation, and parameter-dependent Krawczyk charts; none is implemented or certified in the present work.

The next mathematical step is a validated implicit-function theorem or a parameter-dependent Krawczyk atlas: treat the eight fibre coordinates as parameters and certify a unique transverse solution varying over a declared fibre neighbourhood. The target mathematical structure is

$$F(x, y) = 0, \qquad x \in \mathbb{R}^{16},\ y \in \mathbb{R}^8,$$

and the goal is to prove that a unique transverse solution $x = x(y)$ exists and is regular over a declared y-region, not to prove a single global isolated root.

10 Reproducibility

The complete v1.3 exact-root pipeline freezes the cohort, transverse bases, preconditioners, and Krawczyk radius, excludes runtime metadata from hashed proof objects, and produces byte-identical formal certificates in two complete runs. This byte-identical claim applies to the formal Arb/Krawczyk proof artifacts, not to the ordinary float64 candidate-generation and held-out prospective $\mathcal{G}_4$ diagnostics. Every ordinary held-out floating-point outcome used by the formal certificate is also verified to lie inside its formal ball as an integrity diagnostic. A clean-environment reproduction in fresh runtimes from tag `v0.3.1` (commit `284974c9f6b952f4e114c8c5bdc9c2c299c4065c`) passed with no vendor account. Both clean-environment runs were executed by the author within the same research pipeline; they are an internal reproducibility check, not an independent third-party reproduction. Hashes, scripts, dependencies, and run records are given in Appendices B and C.

11 Conclusion

The main result is a computer-assisted exact-root finite-error ordering theorem: on the frozen twelve-path cohort, strict Krawczyk inclusions certify twelve locally unique roots exactly matched in projective output state and first projective response, and direct 192-bit outward-rounded Arb propagation of the certified root boxes certifies the complete frozen ordering for all 66 unordered path pairs.

The mechanism behind the ordering is geometric. Exact projective first-response matching makes the quadratic loss form common, so the first possible discriminator is the quartic response tensor, whose covariant contraction with the noise moment organizes the comparison: it certifies 34/66 pairs alone, and a zero-error jet through order thirty certifies 52/66 with no reversals as a computationally distinct mechanism certificate. On architecture-specific executions of the separate twenty-path prospective protocol, the frozen quartic contraction predicts the held-out ranking above the predeclared $\rho_{\mathrm{Spearman}} \geq 0.95$ gate; the exact correlation and named top path are platform-dependent. This threshold-level evidence is an application of the same structure, not a premise of the theorem.

The certificate is conditional on the declared serialized model and error functional. The next mathematical step is a parameterized Krawczyk or validated implicit-function continuation over the fibre coordinates; scalability, open-system, and hardware extensions each require new certificates, not extensions of the present one.

A Interval-arithmetic certificate construction

For completeness we collect the construction parameters of the formal certificates. All formal objects are computed in 192-bit outward-rounded Arb/ACB ball arithmetic [7]: every phase-dependent trigonometric factor, Hamiltonian matrix exponential, state propagation, overlap, infidelity, and six-point mean is enclosed. For each declared path, the Krawczyk computation uses a point preconditioner given by a numerical inverse of the midpoint Jacobian, a common transverse box radius of 3×10^{-12}, an interval enclosure of the Jacobian over the full box, and the strict inclusion test $K_k(X) \subset \mathrm{int}(X)$; the projective chart denominator ψ_0 is verified to exclude zero on every accepted box. The order-thirty mechanism certificate extracts Taylor coefficients by a 64-point outward-rounded Cauchy discrete Fourier transform; the discrete-transform alias is enclosed by the path-uniform Dyson bound $e^{K_a} K_a^{64}/64! < 10^{-40}$ on every physical error axis, and the mean six-axis analytic even Dyson tail after order thirty is enclosed by 2×10^{-11} (recorded mean tail $1.2342576 \ldots \times 10^{-11}$), while the largest individual-axis tail is below 4×10^{-11} (recorded maximum $3.7027728 \ldots \times 10^{-11}$). As an integrity diagnostic, every ordinary held-out floating-point outcome is verified to lie inside its corresponding formal ball; this diagnostic is not part of the theorem proofs.

B Certificate identifiers and integrity diagnostics

Rounding and hash conventions. Throughout, outward rounding is used: certified decimal *lower* bounds are rounded downward and *upper* bounds upward, so the reported interval always contains the exact value; a certified *negative* upper bound is rounded toward zero only in the direction that still bounds the true value from above. Consequently the displayed minimum margin $\Delta_{\min} = 2.50 \times 10^{-5}$ is a downward rounding of the certified value $2.505874 \ldots \times 10^{-5}$, and the displayed utilization bound 0.866 is an upward rounding of the computed maximum $0.8653915 \ldots$, so the quoted bound direction-safely exceeds the certified worst utilization. An upper-bound quotation (rather than a truncated equality) is the correct form here: the certified content of the inclusion theorem is the strict inclusion itself, for which only a valid upper bound below 1 is invoked. Certificate identifiers use SHA-256 over a canonical JSON serialization (UTF-8, keys sorted, minimal separators, no insignificant whitespace); runtime metadata is excluded from every hashed proof object. The mapping from each hash to its file is: cohort hash → `results/exact_fibre_krawczyk/cohort.json`; Krawczyk protocol/certificate → `results/exact_fibre_krawczyk/protocol.json` and `krawczyk_certificate.json`; exact-root ordering protocol/certificate → `results/exact_root_ordering/protocol.json` and `exact_root_ordering_certificate.json`.

The frozen twelve-path cohort hash is

`9942af610c1e9499a10e989e4ae069e3`
`912d58eec4fd3ad1d3d39947c232c356`.

The deterministic protocol and certificate hashes are:

- Krawczyk protocol:
  `a19d01ec77a783be4e3858dd92ca080a`
  `a9fde2b09b77541b0e1ad7f392ff3124`
- Krawczyk certificate:
  `fc3ad3823597bd5281b7b9292f227f72`
  `3c2b3a9a8aa4e42c9f0353921b4d8fec`
- Exact-root ordering protocol:
  `f51ecfd2e4572f271e276cae18697005`
  `baa3f0ec3f3ab94f98f919cd50c3f844`
- Exact-root ordering certificate:
  `03c4db75cb80a149ce1a61e11c227039`
  `f9bd79c3f1304b3defa311a491ea2d1e`.

Remark 5 (Role of P0/P1/P2). The frozen point preconditioners used in Theorem 1 are certified nonsingular by an independent 256-bit outward-rounded Rump–Neumann regularity certificate (`results/audit_closure/p0_preconditioner_certificate.json`): $\|I - R_k C_k\|_\infty < 1$ for all twelve paths (12/12 nonsingular). This is a preconditioner-regularity certificate, not a second full interval ordering proof. Three audit roles are distinguished by release version. The **v0.3.1 formal proof snapshot** already contains the strict Krawczyk inclusions that are the theorem evidence; it needs no further certificate. The **P0/P1/P2** artifacts belong to the later **v0.3.2 audit-closure supplement**: P0 is an independent redundant recheck of production preconditioner regularity, P1 is an independent high-precision model rebuild, and P2 is an analytic/mutation test. They strengthen auditability but neither replace, modify, nor form part of the original v0.3.1 frozen proof snapshot.

Table 3: Numerical integrity diagnostics from representative Colab runs.

Diagnostic	Value
Fourth-tensor independent components	15
Maximum tensor recovery relative residual	4.63×10^{-15}
Direct coordinate-refit relative error	1.08×10^{-14}
Invariant-contraction relative error	2.36×10^{-15}
Quadratic-coefficient relative spread	1.09×10^{-7}
Arb precision	192 bits
Cauchy points	64
Taylor order	30
Formal quartic pair coverage	$34/66 = 51.52\%$
Exact-root order-30 mechanism coverage	$52/66 = 78.79\%$
Exact-root direct Arb ordering coverage	$66/66 = 100\%$

C Scripts and clean-environment reproduction

The principal standalone scripts are:

```
pasqal_two_atom_G4_standalone_colab.py
pasqal_L3_L4_standalone_colab.py
pasqal_L4_order30_standalone_colab.py
pasqal_L4_formal_arb_standalone_colab.py
pasqal_L4_exact_fibre_krawczyk_standalone_colab_v1_2.py
pasqal_L4_exact_root_ordering_standalone_colab.py
pasqal_L4_reproducible_certificate_v1_3_colab.py
```

The last script produces all formal proof objects. The formal standalone script pins `python-flint==0.8.0`. NumPy and SciPy provide numerical diagnostics, while the v1.3 proof consumes the frozen decimal cohort and uses Arb/ACB for the formal interval certificate. The "v1.3 pipeline" denotes the formal reproduction script generation used by the certificate snapshot frozen at repository tag `v0.3.1`; it is distinct from manuscript version v1.2 and from the audit-closure tag `v0.3.2`.

The two clean-environment Colab executions of Section 10, started from the public v0.3.1 tag, completed in 52.6 s and 107.9 s respectively; each run covers dependency installation, repository SHA-256 verification, reference-artifact validation, regression tests, and the two-run formal recomputation.

Code and Data Availability

All code, frozen inputs, formal certificates, and reproduction scripts are available at

https://github.com/papasop/quantum-control-geometry

Four distinct objects must not be conflated. The **v0.3.1 formal proof snapshot** (commit `284974c9f6b952f4e114c8c5bdc9c2c299c4065c`) fixes the formal certificates used in this work. The **v0.3.2 audit closure** adds the supplementary P0/P1/P2 audits described in Remark 5 without altering the v0.3.1 proof. The **current manuscript** is a textual revision synchronized on the repository's `main` branch; its wording is aligned to the frozen v0.3.1 proof objects but does not change them. Separately, any previously archived record DOI refers to an older PDF, not to the present text. No PASQAL account or cloud credentials are required.

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The author declares no competing interests.

Author Contributions

Y.Y.N.Li conceived the study, developed the theoretical framework, implemented the numerical and formal audits, analyzed the results, and wrote the manuscript.

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